

The first nontrivial complex finite Grothendieck dimension is three

Automated mathematics run `fa_banach_001`
Lane 4, `agent_lane_04`, model GPT5.6

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Status. Candidate full solution, likely valid, pending human review. The proof is exact; numerical optimization is used only as a sanity check.

1 Source question

Yu, Duan, and Xu define the complex finite Grothendieck constant K_d as the least K for which the unit ball $\mathbb{S}_d$ of $d \times d$ Schur multipliers is contained in $K \operatorname{conv}(\mathbb{D}_d)$, where $\mathbb{D}_d$ consists of rank-one Schur multipliers. They recall that $K_2 = 1$ and ask for the first integer d such that $K_d > 1$ [1, p. 12]:

“It seems, however, that the first integer d for which $K_d > 1$ is still unknown.”

proved independently by Davie and the third named author that $K_2 = 1$ (see the remark at the end of Chapter 5 of Ref. [27]). It seems, however, that the first integer d for which $K_d > 1$ is still unknown. This problem is related to the characterization of the extreme points of $\mathbb{S}_d$. Indeed, $K_d > 1$ is equivalent to the existence of extreme points Φ_S of $\mathbb{S}_d$ that are not of the form $S = [\alpha_k^* \beta_j]$ for some complex numbers α_k and β_k with $|\alpha_k| = |\beta_k| = 1$ for all k . Thus such extreme points exist for large d . It would be interesting to characterize the extreme points of $\mathbb{S}_d$ in the spirit of Refs. [2] and [7].

Figure 1: The source question on PDF page 12 of arXiv:1201.1172.

The AQBC conjecture flagged by the deterministic scan is not the target of this packet: it is disproved in the same source paper. The first-dimension question above is a separate, genuinely open statement in the source.

2 Definitions and separation criterion

We use inner products linear in the second variable. The factorization description of Schur multiplier norm gives

$$\mathbb{S}_d = \{S = (s_{ij}) : s_{ij} = \langle \xi_i, \eta_j \rangle, \|\xi_i\|, \|\eta_j\| \leq 1\},$$

where the vectors may lie in any finite-dimensional Hilbert space. The rank-one set is

$$\mathbb{D}_d = \{(\overline{\alpha_i} \beta_j) : |\alpha_i|, |\beta_j| \leq 1\}.$$

For a coefficient matrix $A = (a_{ij})$, define its vector and scalar support values by

$$V(A) = \sup_{\|\xi_i\|, \|\eta_j\| \leq 1} \left| \sum_{i,j} a_{ij} \langle \xi_i, \eta_j \rangle \right|,$$

$$C(A) = \sup_{|\alpha_i|, |\beta_j| \leq 1} \left| \sum_{i,j} a_{ij} \overline{\alpha_i} \beta_j \right|.$$

Thus $V(A)$ and $C(A)$ are the support functions of $\mathbb{S}_d$ and $\text{conv}(\mathbb{D}_d)$, respectively. Consequently,

$$V(A) > C(A) \implies K_d > 1. \quad (1)$$

This is simply strict separation of the two compact convex sets.

3 Proof intuition

The witness is a small Hermitian matrix with entries among $\{1, -1, i, -i\}$. A carefully chosen rank-two Gram matrix makes its three vector row sums have squared norms $8/3, 6, 6$, producing the exact vector value $8\sqrt{6}/3$.

For scalar phases, optimizing the three row phases leaves only two column phases. Cauchy–Schwarz and one weighted square reduce their total value to a Hermitian three-cycle quadratic form. Fixing one phase collapses the quadratic maximization to the elementary function $a + \sqrt{30 - 10a}$. Its upper bound $11/2$ is sharp for the quadratic relaxation, but the corresponding phases cannot satisfy equality in the weighted square. This incompatibility makes the scalar bound strict.

4 Full solution

Theorem 1. *For the complex finite-dimensional constants in [1], one has $K_3 > 1$. Hence the first integer d for which $K_d > 1$ is $d = 3$.*

Proof. The equality $K_1 = 1$ is immediate from the definitions, and the known assertion $K_2 = 1$ is recalled in the source passage above. It is therefore enough to prove $K_3 > 1$.

Consider

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & -i \\ 1 & i & -1 \end{pmatrix}. \quad (2)$$

We first lower-bound $V(A)$. Let η_1, η_2, η_3 be unit vectors with Gram matrix

$$G = \begin{pmatrix} 1 & (-5 - 7i)/18 & (-5 + 7i)/18 \\ (-5 + 7i)/18 & 1 & (7 - 15i)/18 \\ (-5 - 7i)/18 & (7 + 15i)/18 & 1 \end{pmatrix}. \quad (3)$$

This matrix is positive semidefinite: its eigenvalues are

$$0, \quad \frac{3}{2} - \frac{\sqrt{179}}{18}, \quad \frac{3}{2} + \frac{\sqrt{179}}{18}.$$

It therefore is the Gram matrix of three unit vectors in $\mathbb{C}^2$. For the i th row of A , put $w_i = \sum_j a_{ij}\eta_j$. Direct multiplication using (3) gives

$$\|w_1\|^2 = \frac{8}{3}, \quad \|w_2\|^2 = 6, \quad \|w_3\|^2 = 6. \quad (4)$$

Taking $\xi_i = w_i/\|w_i\|$ yields

$$V(A) \geq \sum_i \|\xi_i\| = \sqrt{\frac{8}{3}} + 2\sqrt{6} = \frac{8\sqrt{6}}{3}. \quad (5)$$

We now prove that the scalar support value is strictly smaller. Since a linear functional on each unit disk is maximized on its boundary, and a common phase may be normalized, write the three column phases as $1, u, v$ with $|u| = |v| = 1$. Optimizing the row phases gives

$$C(A) = \max_{|u|=|v|=1} (r_1 + r_2 + r_3),$$

where

$$\begin{aligned} r_1 &= |1 + u + v|, \\ r_2 &= |1 - u - iv|, \\ r_3 &= |1 + iu - v|. \end{aligned}$$

Set $r = r_1$ and $Q = r_2^2 + r_3^2$. First,

$$r_1 + r_2 + r_3 \leq r + \sqrt{2Q}.$$

The nonnegative square $(\sqrt{3}r - \sqrt{2Q/3})^2$ gives

$$(r + \sqrt{2Q})^2 \leq 4r^2 + \frac{8}{3}Q, \quad (6)$$

with equality only when $\sqrt{2Q} = 3r$.

A direct expansion shows that the right-hand side of (6) equals

$$28 + \frac{8}{3} \operatorname{Re}((1 + 2i)u + (1 - 2i)v + (3 + 4i)\bar{u}v). \quad (7)$$

Put $a = \operatorname{Re}((1 + 2i)u)$. For fixed u , the real part inside (7) is at most

$$a + |(1 - 2i) + (3 + 4i)\bar{u}| = a + \sqrt{30 - 10a}.$$

Since $a \leq \sqrt{5} < 11/2$, squaring the desired inequality is legitimate, and

$$\left(\frac{11}{2} - a\right)^2 - (30 - 10a) = \left(a - \frac{1}{2}\right)^2 \geq 0.$$

Therefore

$$4r^2 + \frac{8}{3}Q \leq 28 + \frac{8}{3} \cdot \frac{11}{2} = \frac{128}{3} = \left(\frac{8\sqrt{6}}{3}\right)^2. \quad (8)$$

It remains essential to verify strictness, since the quadratic bound itself is sharp. Equality in (8) forces

$$a = \frac{1}{2}, \quad v = \frac{(1-2i) + (3+4i)\bar{u}}{5}. \quad (9)$$

The two unit-modulus solutions of $\operatorname{Re}((1+2i)u) = 1/2$ are

$$u = \frac{(1-2i)(1 \pm i\sqrt{19})}{10}.$$

Substitution into (9) gives, respectively,

$$r^2 = \frac{112 - 24\sqrt{19}}{25} \quad \text{or} \quad r^2 = \frac{112 + 24\sqrt{19}}{25}. \quad (10)$$

On the other hand, if equality also held in (6), then $2Q = 9r^2$; combining this with equality in (8) would give

$$\frac{128}{3} = 4r^2 + \frac{8}{3}Q = 16r^2, \quad \text{hence} \quad r^2 = \frac{8}{3}.$$

Neither value in (10) equals $8/3$. Thus the chain of scalar inequalities can never attain $128/3$. The phase torus is compact, so

$$C(A) < \frac{8\sqrt{6}}{3} \leq V(A).$$

The separation criterion (1) proves $K_3 > 1$, completing the proof. $\square$

5 Verification notes

The script `code/verify_witness.py` checks symbolically that:

- G is Hermitian with the displayed eigenvalues and rank two;
- the three squared norms in (4) are exactly $8/3, 6, 6$;
- the two equality cases give exactly the values in (10), neither equal to $8/3$.

It also numerically maximizes the two-phase scalar expression as a sanity check. The numerical step is not used in the proof.

Potential reviewer failure modes are limited and explicit: the support functional must use the same conjugation convention as the Schur factorization, and strictness requires the final equality-case exclusion.

6 Limitations and novelty check

The theorem determines only the first dimension at which $K_d > 1$. It does not compute the exact value of K_3 , nor does it characterize all extreme points of $\mathbb{S}_3$.

The bounded novelty search covered the run registry and solution indexes, the local parsed arXiv corpus, and arXiv/web searches for the exact first-integer phrase, complex K_3 , and 3×3 extremal Schur multipliers. We also inspected Christensen's recent work [3], which proves necessary rank bounds for general extremal Schur multipliers but does not supply this K_3 separation or answer the source question. No prior answer was located. Novelty confidence is moderate rather than definitive.

7 Human-review recommendation

Recommend focused expert review. Check equations (4), (7), and the equality-case substitution (10). If those three computations and the standard support-function equivalence are confirmed, the source question is fully answered.

References

- [1] N. Yu, R. Duan, and Q. Xu, *Bounds on the distance between a unital quantum channel and the convex hull of unitary channels, with applications to the asymptotic quantum Birkhoff conjecture*, arXiv:1201.1172 (2012).
- [2] G. Pisier, *Grothendieck’s theorem, past and present*, Bull. Amer. Math. Soc. 49 (2012), 237–323.
- [3] E. Christensen, *Extremal Schur Multipliers*, arXiv:2410.20112 (2024; latest inspected version dated March 2026).